

15 | OVERVIEW



Restoring Marine Ecosystems and Human Health: Artificial Reefs in Madagascar

Abstract

Coral reefs feed millions of people, but Madagascar's reefs have lost 30 to 50% of their coral cover since 1998. The Health Impacts of Artificial Reef Advancement (HIARA) project, led by Harvard Chan C-CHANGE with local partners, is testing whether rebuilding reefs in the Bay of Ranobe can restore marine life, strengthen fisheries, and improve the nutrition and health of nearby communities.

Introduction

Coral reefs are nurseries for the ocean, giving young fish and other sea creatures a safe place to grow. When reefs decline, fish populations fall, and so does a key source of nutrient-rich food. Worldwide, an estimated 1.39 billion people are vulnerable to worsening nutrient deficiencies because of declines in fish. In southwestern Madagascar, the reefs of the Bay of Ranobe, part of the Toliara Barrier Reef, are stressed by overfishing, unsustainable tourism, and global coral bleaching. Sediment from deforestation can smother corals and trigger algal blooms that deplete oxygen. Local communities depend heavily on the sea, so a struggling reef means struggling families.

How Do Scientists Study This?

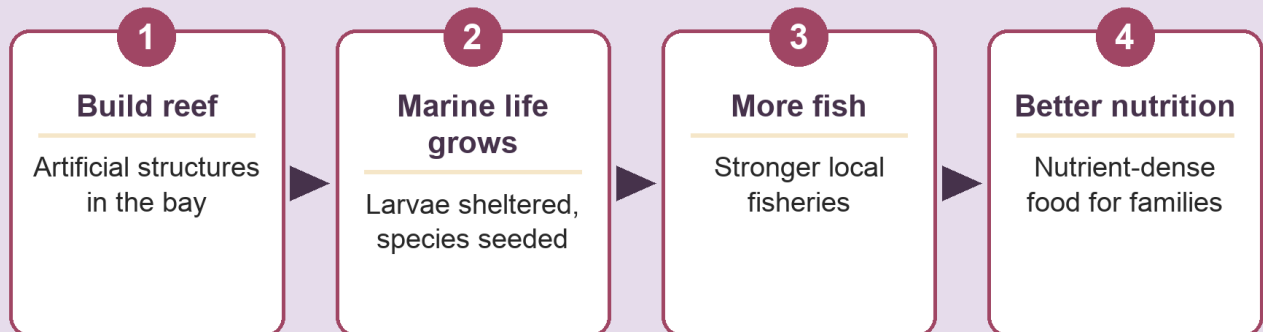


FROM THE ARTICLE

Project location: Bay of Ranobe, Madagascar. Map: Harvard Chan C-CHANGE.

HIARA combines ecology, fisheries science, and public health. The team uses two new reef-building technologies. The first uses small structures that gently collect sea creatures from healthy reefs and transfer them to newly built reefs that need to be seeded with diverse marine life. The second uses artificial reef designs that attract sea creatures in their larval stage and protect them as they grow. Local community members help design, deploy, and monitor the reefs. Researchers then collect a wide range of data, including reef and water quality surveys, fish catch assessments, and health measures such as diet, blood pressure, mercury levels, body measurements, and symptoms of depression and anxiety.

From reef to health



Study the Process

Build reefs

New artificial reef structures are designed and placed in the Bay of Ranobe with local communities.

Seed with life

Sea creatures are transferred from healthy reefs, and designs attract and shelter larvae.

Track reef and people

Scientists monitor reef health, fish catch, diets, and community health over time.

What Did They Discover?

The project is testing whether a healthier reef leads to healthier people. By bringing in more kinds and greater numbers of sea life, the artificial reefs aim to boost local fisheries and give families better access to nutrient-dense foods. Because the data span ecology, fisheries, and health, researchers will be able to trace the whole chain from coral to diet to well-being. The approach is designed in southwestern Madagascar but could apply to more than 100 countries with coral reefs. The results will provide evidence for conservation groups and public health institutions around the world considering interventions that benefit both people and nature.

Reefs and nutrition

30–50%

decline in Madagascar coral cover since 1998

1.39B

people at risk of nutrient deficiency as fish decline

100+

countries with coral reefs that could apply this approach

Madagascar's reefs lost about 20% of their coral cover in just the last 20 years, showing how quickly these ecosystems are changing.

15 | TAKEAWAYS

Takeaways

Reefs feed people

Healthy coral reefs support fish that provide essential nutrients for coastal communities.

Rebuild with science

New designs collect and shelter marine life to restore damaged reefs.

Communities lead

Local residents help design, deploy, and monitor the artificial reefs.

Key Terms

Coral reef

An underwater ecosystem built by tiny animals called corals, which provides food and shelter for many sea creatures.

Coral bleaching

When stressed corals, often from warm water, expel the algae living in them and turn white, which can lead to their death.

Artificial reef

A human-made underwater structure designed to support marine life and help restore a reef ecosystem.

Larval stage

An early life stage of many sea creatures, when they are tiny and especially vulnerable.

Nutrient deficiency

A lack of essential vitamins or minerals in the diet, which can harm growth and health.

15 | ACTIVITY

Activity

Reef-to-Plate Food Web

Students trace how coral reef health connects to the food and health of coastal families.

Materials: Cards for coral, larvae, fish, fishers, families, sediment, warm water; String; Poster paper

1. Arrange the cards into a chain from coral reef to family dinner table and connect them with string.
2. Add the threats from the lesson (overfishing, bleaching, sediment) and show which links they break.
3. Show where the HIARA artificial reefs would repair the chain.
4. Choose two measurements from the project, one ecological and one health-related, and explain what each would show.
5. Write a short paragraph explaining why a reef project counts as a health project.

References

Harvard T.H. Chan School of Public Health, C-CHANGE. Artificial Reefs to Support Human Nutrition. <https://hsph.harvard.edu/research/climate-health-c-change/research/artificial-reefs-to-support-human-nutrition/>